

The $O(1)$ -State Organism: Seven Measured Foundations for Continuously Living Sequence Models

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Every number in this paper is committed in the public repository github.com/DT-Foss/o1-state with its generating script and raw JSON; the prediction ledger ([analysis/PREDICTIONS.md](#)) was written before the data it scores. This draft is itself part of the public disclosure.

Abstract

Contemporary sequence models are batch artifacts: trained on a frozen corpus, frozen at deployment, and structurally incapable of one thing no scaling law addresses — **they never experience their own future**. We present measured foundations for the alternative: a bounded-state organism that consumes an unbounded stream at constant memory, decides *from its own prediction error* when to learn, what to remember, when to consult an external index, and when to sleep — and whose complete living state is a small, portable, forkable asset. Three measurements carry the case. **(i)** In a pre-registered 40-hour experiment (909.7M streamed tokens, 16/16 integrity checks), the organism’s surprise gate did not merely approach full-gradient training at 25.2% of the gradient tokens — it **beat** it (improvement ratio 1.0091), crossing the prediction’s own registered “embarrassment threshold.” A width curve — corrected mid-flight when its own provenance trail exposed a config-default flip masquerading as a width effect — now runs to $d=1024$ at fixed dose: the improvement ratio reads 0.9708, 0.9892, 0.9777, **1.0092**, the gated arm crossing full-gradient training a second time, on the width axis. The gate rate itself proved a dial, not an emergent: five runs at $q=0.8$ spanning $8\times$ in width land within 0.13pp of one rate. **(ii)** A 1,713,673-parameter organism has now streamed **7.05 billion tokens in one continuous life** at a process memory that never left 0.69–0.83 GB, 0.87B→7.05B of it in a single OS process over roughly twelve days; its one stall self-healed from its own atomic checkpoint. **(iii)** Two live machines on a real network, source never paused: at the one equal chunk count the held-out delta between an ARM and an x86 replica is **0.0, exact**. We further measure: an exactness license swept over four operators and six (chunk, overlap) points — layout decoupling exact to 0.0, gradient exactness gated by warmup *overlap* rather than chunk length; a two-timescale law from three independent experiments; keyed holographic recall through silence with a movable persistence knee; collective memory (one organism’s stored surprises cut another’s shock-forgetting to $0.67\times$); and family-generic transfer of the entire operating mode to a Mamba/S6 parametrization ($0.98\times$). Forty-two predictions were registered before their data; the falsifications — a replay-cost law, a rank hypothesis, two migration thresholds — are reported at the same strength as the confirmations, with the numbers that killed them. We argue these primitives compose into a **reality prediction engine**: a system that lives on streams of the present, deposits predictions about its own future at every horizon, and learns from their arrival — an axis of capability orthogonal to parameter scaling, and one on which batch models cannot compete by construction.

1 The blind spot

A language model trained on a fixed corpus optimizes next-token prediction on the *past*. At deployment its weights are dead; its context window is a purchased illusion of memory; a session’s experience evaporates. Three structural consequences follow. (i) The model cannot *verify* its own predictions — verification requires living until the predicted moment arrives. (ii) It cannot *accumulate* — its state is either bounded-and-overwritten or unbounded-and-unaffordable. (iii) It cannot *move* — the serving process is welded to the hardware that holds its KV cache.

This paper measures the alternative at small scale and full methodological rigor. The claim is not that a 128-dimensional organism rivals a frontier model at language; it does not. The claim is that the **operating layer** every persistent deployed model will need — when to learn, what to keep, how to move, how to sleep, how to verify — is measurable *now*, that its laws are architecture-generic, and that they are already surprising: the learner’s own prediction error turns out to be not merely a sufficient control signal but a *better* teacher than learning from everything.

2 The substrate (prior work, compressed)

The organism runs on GSSM-Selective, a gated bounded recurrence $z_t = \gamma_t z_{t-1} + a_t$ in which Mamba/S6 [3], S5 [8], and LRU [6] are switch restrictions of one affine prefix-scan operator (family reduction measured at $\sim 1e-15$; repository §1). Three prior structural results carry the present work: **(a)** the operator is shift-equivariant with a $\sim 5-8$ -token contraction receptive field, giving perplexity that *improves* to $524,288\times$ the training length (§8) and a billion streamed tokens at flat RSS; **(b)** truncated-BPTT with a detached carried state reproduces full-window gradients exactly — streaming *training* is exact, not approximate, under a condition this draft measures precisely (§6); **(c)** gated readouts have a sharp capacity cliff at load ≈ 1 (slope 1.32 vs. 0.57 linear), which *forces* the two-system split between a bounded live state and a growable external index.

3 Seven foundations

Each is stated over the operator class and anchored by committed measurements. (Full formal statements: FOUNDATIONS.md.)

F1 — The surprise calculus. The learner’s own prediction error suffices as the universal control signal: when to learn (gate), what to remember (spike-selected spans), when to consult (retrieval triggers), when and how much to sleep (dividend-monitored replay), how curious to be. *Anchor*: the 40h result of §4 and its width curve (§5). *Extension (disclosed, first experiment registered)*: a ladder of horizons — deposit predictions about the stream’s own future, score them on arrival, learn at every timescale.

F2 — The exactness license. Bounded contraction makes detach-carry streaming exact and decouples training layout from deployment layout. Swept over four operators $\times$ six (chunk, overlap) points, the license holds with a precisely located boundary: layout decoupling is exact to 0.0 everywhere; gradient exactness is bought by warmup *overlap*, not by chunk length (§6).

F3 — Phase–magnitude separation. In complex bound states, content (phase) is written and never evolves (zero-drive invariance $|\Delta\varphi| \approx 1e-8$, machine precision); persistence (magnitude) decays. Real fillers are *double agents* — they pollute the phase (removable exactly by an eval-time clamp:

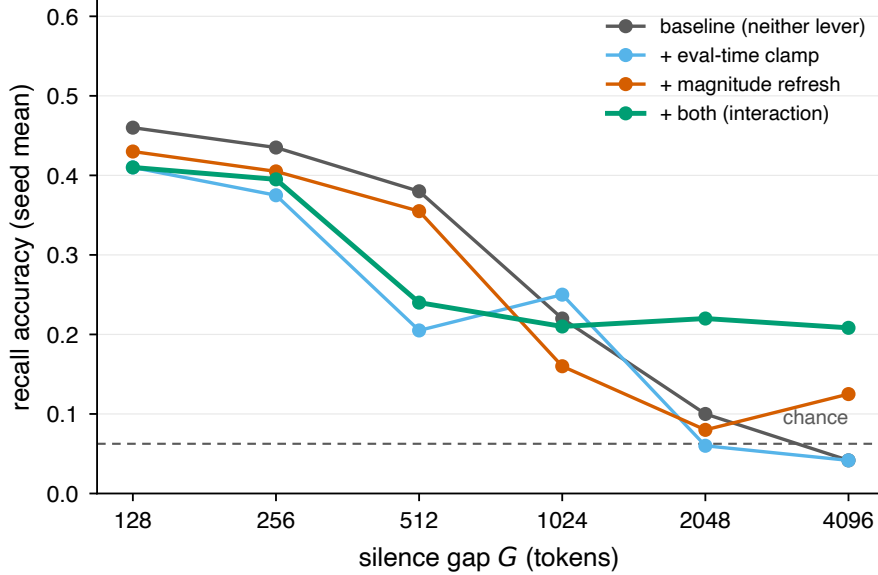


Figure 1: The persistence knee moves with the eval-time levers. Each lever alone decays toward chance beyond $G \approx 512$ – 1024 ; applying both together holds recall at more than $3\times$ chance out to $G=4096$ — an interaction, not a sum. Source: `results/holo_clamp_refresh.json`.

drift 0.0 rad) while *feeding* the magnitude (clamped, channels starve). The recall knee moved $32 \rightarrow 256 \rightarrow 512 \rightarrow 2176$ mean / 4096 end-of-range across theory-led interventions, the last requiring BOTH levers jointly — an interaction, not a sum. Knee *position* is seed-dependent; the intervention *effects* replicate.

F4 — The two-system law. The gated-readout cliff (rank-1-per-channel binding, $D_{\text{eff}} \approx 1.02$ measured against a closed-form bound) forces unbounded accumulation into an external index. Runtime consultation lifts recall far past state capacity: at 16 pairs, 0.150–0.158 unaided $\rightarrow$ 0.310–0.512 consulted, dose-dependent (a full gate roughly doubles the half gate) with the random-injection control at 0.100–0.143. Training *with* stochastic consultation makes the read near-perfect at low load — 0.995–0.998 at 2 pairs, against 0.530–0.565 unaided — and still lifts 16 pairs to 0.487–0.505. Held-out-key controls kill the lookup-table explanation: the binding generalizes to never-trained keys.

F5 — Family-generic operating modes. The full POS recipe run on a scan-parameter-matched Mamba/S6 configuration transfers at $0.98\times$ (three seeds, two CPU architectures) — and the house operator beats S6 head-to-head (+0.13 to +0.16 nats) at parity. The calculus attaches to the operator class, not to one network.

F6 — Train short, deploy unbounded. One shift-equivariance symmetry on two axes: sequence length (perplexity improving to $524,288\times$; corpus-as-iterator to 10^9 tokens and beyond) and silence (keyed bindings held through gaps $128\times$ the training horizon). Its honest boundary is measured twice: multi-chunk training is structurally gradient-blind to the write — the cure is to train inside the boundary and let exactness carry the skill out.

Table 1: The 40-hour surprise-gated run vs. full-gradient training.

arm	held-out 8.6588 $\rightarrow$	improvement	gradient tokens
A2 full-gradient	4.7782	3.8806	100%
A3 surprise-gated	4.7430	3.9158	25.17%

F7 — The portable organism. The complete living system — weights, optimizer moments, carried state, gating windows, span store, stream position, RNG — is one atomic ~ 53 MB artifact, and the organs couple through *kilobytes* (spans, reminders, deltas), not activations. Measured: local checkpoint $\rightarrow$ new-process $\rightarrow$ resume is bit-identical; **live mid-stream migration from Apple Silicon to x86 continues behaviorally identically to six decimals** (heldout 6.182391 == 6.182391, every gate decision matched — only the BLAS bit-digest differs, and it does not propagate); the same parity holds *between two live machines on a real network* (§10); a killed replica rejoining through a *shared* span store ends *better* than its never-killed twin (-0.014 : the partner’s outage-window spans are a gift); an outage spent sleeping beats one spent idle ($+0.065$).

4 The 40-hour experiment: the gate beats the firehose

One process, three from-scratch arms on one cloned C4 stream (seed 42), 40.0 hours, 909,676,544 tokens, no restarts, $\text{RSS} \leq 1.094$ GB. A1: forward only. A2: backward on every chunk. A3: backward only when the chunk’s own mean NLL clears the rolling 75th percentile of its own history — no oracle, no second model, no labels, no schedule.

Ratio 1.0091. The pre-registered prediction (P1) put the point estimate at 0.85 with an explicit embarrassment threshold: “*ratio > 1.0 sustained would be a bigger result than the thesis itself.*” It fired. The surprising part is not efficiency — it is that the 75% of chunks the gate skips are, in net, *worth skipping*: selective plasticity beats indiscriminate plasticity at equal data exposure. Every number re-derives from the raw 1.78M-chunk log (16/16 integrity checks).

The twin. At T+24h, A3’s weights were copied into a fresh process with zero carried state. Registered: a warmup tax ($+0.03$ – 0.15 excess surprise, $\geq 1.5\times$ over-gating, hours to converge). Measured: excess 0.0029, post-fork gate rates 0.2521 vs. 0.2522, convergence one chunk after the fork, and the twin finished *ahead* (4.7365). Restarting a living stream is free at the fast timescale — the third independent measurement of the two-timescale law (§9).

The honest negative. 1020 paired index-injection probes: both deltas negative (the index froze at its 20k cap by token 11M; replaying 5–11M-era spans into an $80\times$ -older model disturbs), but injection hurts $3.2\times$ less than random injection (-0.087 vs. -0.279 ; helped 0.35 vs. 0.10) — the mechanism carries information; the provenance was stale. Registered as falsified; the v2 (rolling spike threshold) is specified.

5 The width curve: a two-point law, falsified by its own third point

The natural next question is whether the gate’s advantage survives width. P42 registered it as a scaling axis of the core result, with the point call stated in advance: the ratio at $d=256$ should land at or above the $d=128$ anchor, because selection ought to matter *more* where each gradient is four times as expensive. The same recipe ($q=0.75$, window 500, $B=8$, $K=64$, lr $3e-3$, seed 42, from-scratch, streamed C4) was run with only the width changed, all three points read at a matched $\sim 50\text{M}$ -token anchor.

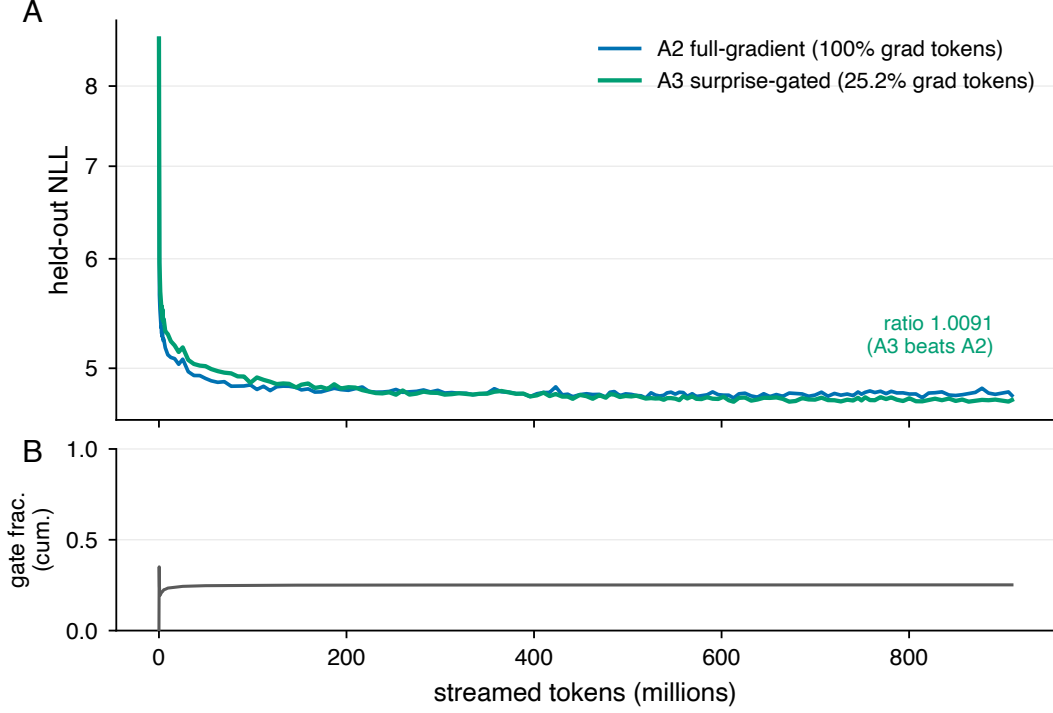


Figure 2: The 40-hour run: held-out loss for A2 (full-gradient) and A3 (surprise-gated), and A3’s cumulative gate fraction, vs. streamed tokens. A3 tracks A2 while training on 25.2% of the gradient tokens and finishes ahead (ratio 1.0091). Source: `results/pos_metrics.jsonl`.

At two points the reading was clean and was reported as such: $0.9729 \rightarrow 0.9953$, the gate law *grows* with width. **The third point falsified it** — the ratio fell to 0.9777 at $d=512$ — and the first version of this section explained the fall by a dosing difference it could observe but not derive: the $d=512$ gate fired at 19.8% where both narrower runs fired at 24.7%, stably from the start, and a forensic hunt for the mechanism (ignition-descent depth, pre-registered as P45) followed. **The provenance trail then closed the case from the other side.** The machine-written run configs record that the $d=512$ run was launched at $q=0.8$ — the runner’s default — where the narrower runs had $q=0.75$ set explicitly. The gap was a config flip, not physics. Two hardening runs at $d=128$ and $d=256$ (seed 43) and the $d=1024$ extension inherited the same default and thereby completed, by accident, the 2×4 q -width grid of Table 2, which separates the two variables cleanly.

The gate rate is a dial, not an emergent. Five $q=0.8$ runs across an $8 \times$ width range land inside 0.1981–0.1994 (0.13pp of scatter, against a 0.02pp seed delta at $d=512$); both $q=0.75$ runs land at 0.2472. There never was a width $\rightarrow$ rate effect. The ignition-depth mechanism of the earlier reading is retracted as a width mechanism — its own forensics had in fact measured the answer: the fraction of window entries above the q_{75} threshold was 0.230 at both $d=256$ and $d=512$, which is precisely what one observes when the $d=512$ run is gating on q_{80} . (The per-chunk ignition traces remain valid measurements of each config’s ignition dynamics, and the early-horizon 2,000-chunk lottery results are untouched.)

What the clean grid shows instead: selection crosses the firehose at $d=1024$. At fixed $q=0.8$ and matched dose (9.90–9.97M gradient tokens), the improvement ratio reads $0.9708 \rightarrow 0.9892 \rightarrow 0.9777/0.9758$ (two seeds agree; the $d=512$ valley is real) $\rightarrow$ **1.0092**: the gated arm ends 0.030 nats *better* than the full-gradient arm on one fifth of the gradient tokens — the same crossing the 40-hour run produced on the token axis (ratio 1.0091 at 909.7M tokens, §4), now reproduced on

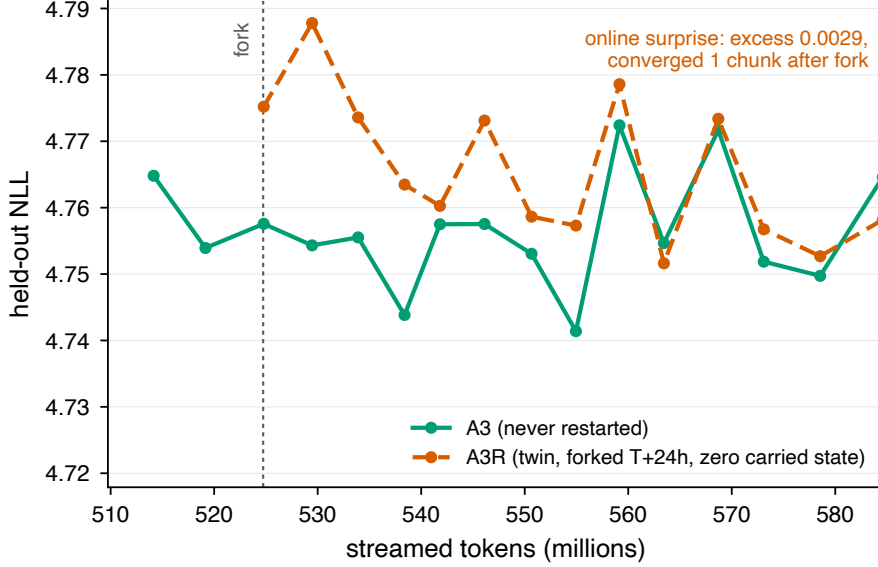


Figure 3: The twin, zoomed to the fork. A3R (forked at T+24h with zero carried state) opens ≈ 0.02 nats above A3, crosses into A3’s own fluctuation band within a few evaluations, and finishes ahead (4.7365 vs. 4.7430). On the registered online-surprise criterion the restart is free: excess 0.0029, post-fork gate fractions 0.2521 vs. 0.2522, convergence one chunk after the fork. Source: `results/pos_metrics.jsonl`, `results/pos_summary.json`.

the width axis at 50M. The $d=128 \rightarrow 256$ rise replicates at both q levels. Per-token efficiency at fixed dose *falls* monotonically with width ($0.3696 \rightarrow 0.3629 \rightarrow 0.3547/0.3467 \rightarrow 0.3305$): the earlier “selection gets cheaper with width” reading compared $q=0.75$ doses against a $q=0.8$ dose, and a higher quantile doses fewer, more selective tokens at any width.

The determinism results are untouched by the correction. An exact repeat of the $d=512$ run (11 days later, a different process, one mid-run stream reconnect) reproduces *all* 97,657 chunks identically on every recorded field — same-seed variance at 50M is zero, and the run-vs-run forks trace to measurement-cell co-load, not to the calculus. A second seed at $d=512$ lands at 0.1986 against 0.1984, with the full profile reproducing (improvement ratio 0.9758 vs. 0.9777; efficiency 0.3467 vs. 0.3547 per million gradient tokens). The 50M cumulative rate is deterministic per (seed,

Table 2: The gate-law width curve on the corrected 2×4 q -width grid, all points at a matched ~ 50 M-token anchor. At fixed $q=0.8$ the dose is matched (9.90–9.97M gradient tokens) and the improvement ratio crosses 1.0 at $d=1024$; the gate rate tracks q alone. Sources: `results/gate_low_width_curve_q08.json`, `results/gate_low_width_curve.json`.

d_{model}	seed	q	gate fires	improvement ratio	improvement / M grad-token
128	42	0.75	24.72%	0.9729	0.2969
256	42	0.75	24.72%	0.9953	0.2937
128	43	0.80	19.87%	0.9708	0.3696
256	43	0.80	19.94%	0.9892	0.3629
512	42/43	0.80	19.84/19.86%	0.9777 / 0.9758	0.3547 / 0.3467
1024	42	0.80	19.81%	1.0092	0.3305

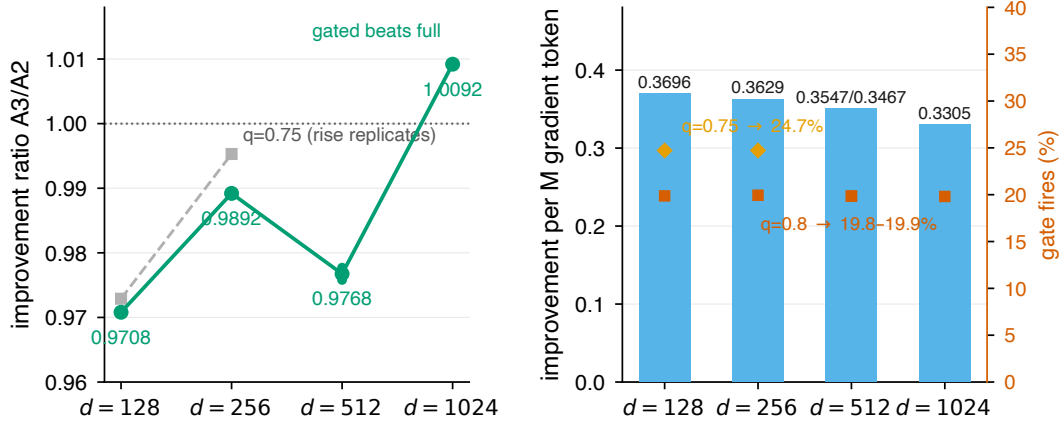


Figure 4: The gate-law width curve on the corrected grid. *Left:* at fixed $q=0.8$ and matched dose the improvement ratio crosses 1.0 at $d=1024$ — the gated arm beats full-gradient training outright; the $q=0.75$ pair (gray, dashed) replicates the $d=128 \rightarrow 256$ rise; both $d=512$ seeds are shown. *Right:* per gradient token spent at fixed dose, efficiency falls with width, and the gate rate separates by q alone — 24.7% at $q=0.75$, 19.8–19.9% at $q=0.8$, across all widths. Source: `results/gate_law_width_curve_q08.json`.

q , width, recipe, stream), and per the grid its width and seed dependence at fixed q is nil. (The $q=0.75$ line rests on one seed per width; its seed-43 pair is running.)

One control on the reading: A2 itself degrades with width at this fixed token budget ($4.9161 \rightarrow 5.0363 \rightarrow 5.0898 \rightarrow 5.4464$), exactly as expected — wider models need more tokens, so the 50M anchor sits earlier in a wider model’s training. The anchor measures selection efficiency, not model quality, and no claim about the latter is made from these runs.

6 The exactness license, swept

F2 is the load-bearing license: if detach-carry streaming were merely approximate, every downstream claim about living training would inherit an uncontrolled error. v0.1 anchored it on one operator at one (chunk, overlap) point. A license with no measured failure mode is a claim, not a law, so the sweep runs four operator configurations — the selective scalar scan, the complex holographic scan, the same scan under the $\tanh\text{-}m$ readout, and a phase-off control — across six (chunk, overlap) points, including chunk sizes deliberately below the stated receptive field.

Two separate statements come out, and only one of them is unconditional. **Layout decoupling is exact to 0.0** — at every operator and every chunk size down to 16, with pure detach-carry (overlap 0), the chunked and full-sequence forward layouts agree to *max-abs-delta* 0.0 exactly. Training layout and deployment layout are interchangeable, full stop. **Gradient exactness is gated by warmup overlap, not by chunk length.** With overlap 16 the gradient cosine is 1.000000000000 at relative error $\sim 5\text{e-}7$; with overlap 0 the relative error runs $5\text{e-}3$ to $2\text{e-}1$ and the worst cosine over the whole sweep is 0.9762 (the complex holographic scan at chunk 16). Chunk length is not what buys exactness — a 128-token chunk with no overlap is worse than a 64-token chunk with overlap 16. The condition is that the carried state be re-warmed past the contraction receptive field before gradients are scored, which is precisely what the $\sim 5\text{--}8$ -token field predicts and what the ≥ 16 -token overlaps deliver.

This supersedes the single-point anchor of v0.1 and states F2 with its boundary attached: the

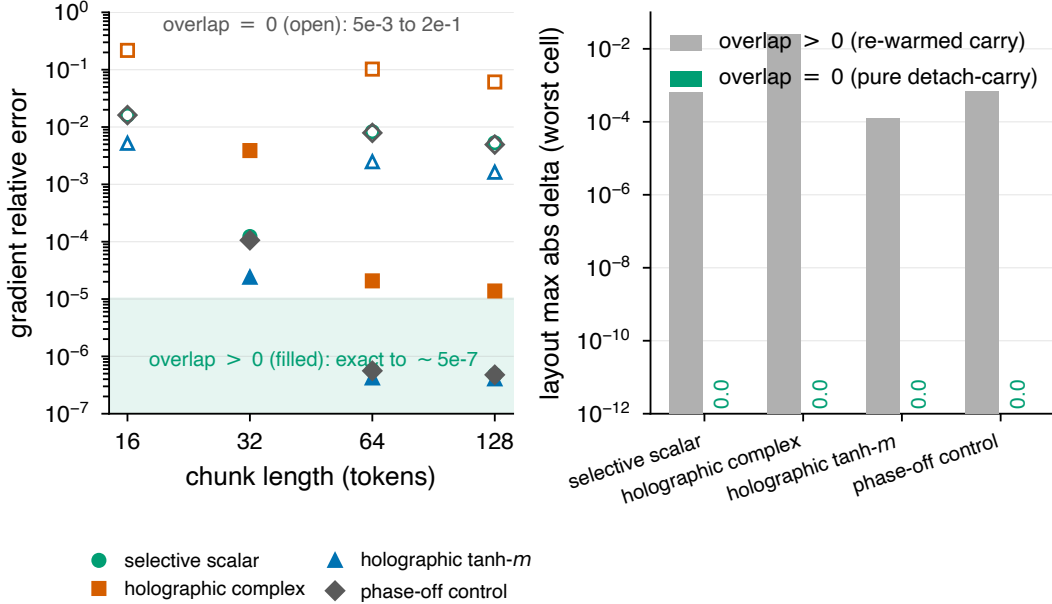


Figure 5: The exactness license, swept over four operators $\times$ six (chunk, overlap) points. *Left*: gradient relative error. Filled markers ($\text{overlap} > 0$) sit in the exact band at $\sim 5e-7$ regardless of chunk length; open markers ($\text{overlap} = 0$) run $5e-3$ to $2e-1$. Exactness is bought by overlap, not by chunk size. *Right*: the layout decoupling is exactly 0.0 with pure detach-carry at every operator, including chunks below the receptive field. Source: `results/f2_equivalence_sweep.json`.

license is real, it covers every operator we run, and it has an operating condition that is measured rather than assumed.

7 Seven billion tokens in one life

A 1,713,673-parameter organism has learned online through **7,053,926,400 streamed tokens in one continuous life** — no dataset stored, no epochs, no growing context — at a process memory that never left the band **0.69–0.83 GB**. The stretch from 0.87B to 7.05B tokens ran in a *single OS process* for roughly twelve days with zero interventions.

The one stall in that life is part of the result rather than an exception to it. On 2026-07-24 an upstream stream hang killed the process at 869,734,400 logged tokens; it resumed from its own atomic checkpoint at 869,683,200 — **51,200 tokens replayed, roughly three minutes of stream lost** — and continued for another six billion tokens without further intervention. The organism healed itself out of its own snapshot, which is exactly what F7 is for.

What this artifact does and does not claim. Over the logged stretch the loss EMA moves $4.25 \rightarrow 4.169$. At 1.7M parameters the model is at its capacity floor, and we state the consequence plainly: this measures the **constancy of resources over experience**, not continued learning. The cost of having lived through seven billion tokens is zero additional memory — that is the claim, and it is the one no batch model can make at any parameter count, because the quantity it holds constant is not a hyperparameter but a lifetime. (A further honesty note the artifact carries: the 0 $\rightarrow$ 512M stretch of this life predates the oldest retained log and is documented only by the run’s status artifact and its start lines; the plotted series therefore begins at 512M tokens.)

The 7-billion-token life: constant memory across the whole of experience
1.7M-parameter organism, online learning on a live C4 stream — no dataset, no epochs, no growing context; one stall, self-healed

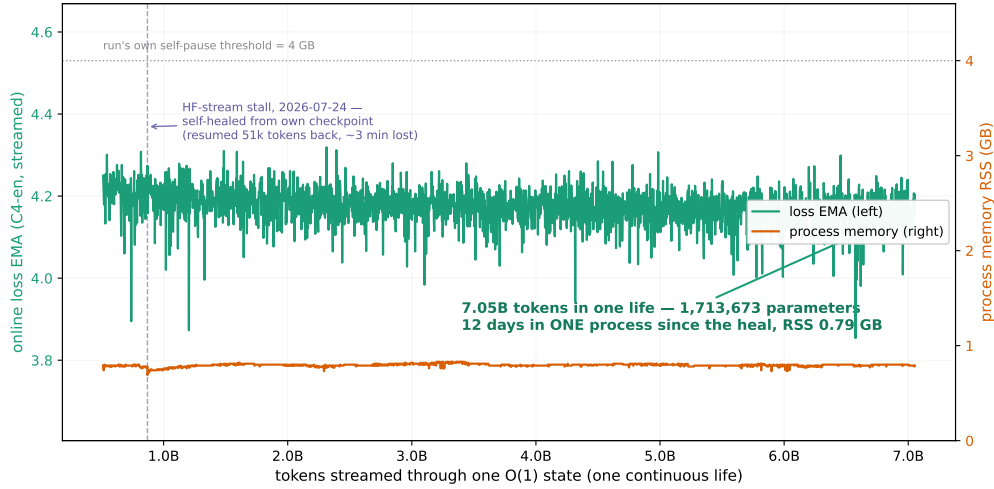


Figure 6: The 7-billion-token life. Loss EMA (left axis) and process RSS (right axis) against tokens streamed through one $O(1)$ state. Memory stays inside 0.69–0.83 GB across almost four orders of magnitude of experience; the marked stall is the single interruption, self-healed from the run’s own atomic checkpoint. Sources: `results/lifetime_7b_series.json`, `results/lifetime_7b_curve.json`.

8 Length, re-anchored

The strongest length statement in this program is not flatness. Using the $O(1)$ recurrent forward with chunked streaming (chunk 8192, overlap 128 $\gg$ the receptive field), a model trained at $T=32$ was evaluated on a single effective sequence of **16,777,216 tokens** — $524,288 \times$ **the training length** — at a peak RSS moving only $1.90 \rightarrow 2.48$ GB, and the perplexity *improves monotonically the whole way* ($\times 1.000$, $\times 0.896$, $\times 0.886$, $\times 0.825$, $\times 0.803$). The model is integrating causal context across the distance as a noise filter; it is not merely failing to crash.

The clean causal ablation says what the wall actually is. At $256 \times$ the training length, the position-free Selective arm sits at $\times 0.973$ while the *identical* architecture with a sinusoidal positional encoding degrades to $\times 4.23$ and the ungated Pure arm explodes to $\times 11.25$. The only difference between the first two arms is the positional encoding. **The wall is the position term, not the length** — and both ingredients are necessary, since removing the gate breaks the bounded state on its own.

9 The two-timescale law of operation

Three independent experiments give the same law. (1) The twin (§4): zero carried state costs nothing. (2) A snapshot cross-matrix (weights at 359M tokens $\times$ states from 128M/240M/359M/zero/shuffled): every arm converges to the native trajectory within 4 chunks — 256 tokens, the receptive-field scale; a cold start is not worse than the carried state. (3) The beacon: a *stored bit* in a write-once-freeze carrier channel (measured at $\gamma = 0.9998$ in the gap, input gate 0.0133 there against 0.5652 at the beacon) survives a d64 $\rightarrow$ d128 widening surgery at recall 1.000 and survives a weight swap across $2 \times$ training distance at recall 1.000 through 512 tokens of silence, against a zeroed-state control at chance (0.475–0.54) — by redundant encoding (the carrier’s channel address can relocate; the read still lands).

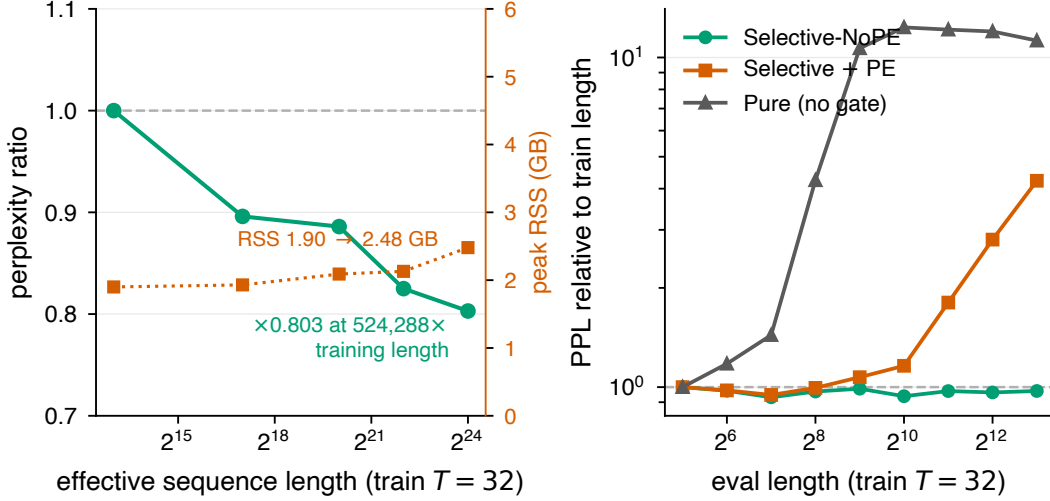


Figure 7: Length, re-anchored. *Left*: perplexity ratio and peak RSS against effective sequence length — $\times 0.803$ at $524,288 \times$ the training length, improving monotonically, at 1.90→2.48 GB. *Right*: the control. Position-free Selective holds $\times 0.973$ to $256 \times$; the same architecture with a positional encoding degrades to $\times 4.23$ and the ungated arm to $\times 11.25$. Sources: `results/scale_to_a_million.json`, `results/length_extrap_v2_extreme.json`.

The law: the fast path forgets by design — so weight updates, model growth (function-preserving widening: surgery equivalence $6.7e-6$, no transient, growth beats restart by $+0.13$ to $+0.17$ nats, $3 \times$ replicated), and migration are *safe live operations*; everything durable lives in slow carriers, weights, and the index — exactly the layer the organism’s own calculus (F1, F4) manages. Operationally this is the answer to a question every persistent-model deployment will face: *what must be preserved across an update?* Measured: almost nothing — if the architecture separates its timescales.

10 Two live machines, one organism

v0.1 reported cross-ISA migration as a checkpoint-and-resume operation. The stronger staging runs it *live*: organism A streams continuously on an Apple Silicon machine and is **never paused**; organism B rises on an x86 server from A’s snapshot transferred over real `scp/ssh` and chases A’s stream position by deterministic replay. At the one chunk count where both organisms stood (560), the held-out delta is **0.0** — **exact**: ARM 6.199778 against x86 6.199778, reproduced across two independent stagings.

The limit of the claim is structural and we state it as the artifact does. **Parity is defined only at equal chunk counts.** B on 16 server cores outruns A on the Mac and overtakes it; at B’s 5480 chunks A had never reached that position (its maximum was 3000), so no comparison exists there — and a comparison at unequal stream positions would not be a parity check at all. Multi-cycle parity therefore requires rate-matched machines. The single-cycle number is the measured one, and it is exact.

Table 3: P39 (a)/(c) measured in situ at the production cadence ($d=128$, $B=8$, $K=64$ — the exact 40h configuration). Source: `results/p39_two_machine_scored.json`.

machine	cores	(a) gap ratio	threshold	(c) CPU ratio B/A	threshold
Mac (ARM)	10	7.085	< 3.0	2.219	< 1.0
beast (x86)	16	5.603	< 3.0	2.039	< 1.0

11 The replay-cost law: two falsifications and the mechanism behind them

P39 registered stop-free migration with four checks. Check (b) — B reaches state parity with A while A never pauses — passes at every cadence and on every machine tried, including the cross-ISA staging above. Checks (a) and (c) are **falsified as registered**, on two independent machines.

The obvious explanation was tested and refuted. If (c) failed because A and B contend for cores on one machine, then giving B its own cores among sixteen should fix it; it moves the ratio by 0.18, not by the $2\times$ it would need. The raw CPU seconds locate the cause instead: **B spends 45.4 s replaying the chunks A produced with 22.3 s of CPU** — replay costs approximately **twice** live streaming per chunk, on both ISAs, contended or not. Check (c) as registered asks replay to run *faster* than production, and nothing in these two runs suggests it can. That is a property of catch-up replay, not of any machine, and it is the standing constraint on continuous replication: a replica cannot catch up on a machine no faster than its source.

A second, methodological lesson rides along. Earlier ratios for the same two checks were inflated by toy chunk weight — 17.935 and 3.789 at $B=4/K=32$, roughly halving to 7.085 and 2.219 once a chunk carries production work. The mechanism is arithmetic: snapshot size scales with the model, not with chunk weight, so raising chunk weight shrinks the snapshot cost measured in chunk slots ($7.94 \rightarrow 2.59$ slots). The cadence hypothesis is therefore half confirmed and half falsified, and both halves stand: toy cadence really did inflate these ratios, and cadence was not the whole story — a residual survives that no cadence correction removes.

12 The organism is social and collective

Organism A streams a C4+code mix, storing its surprise spans. Organism B, facing a code shock, replays A’s shared spans and forgets **$0.67\times$** of what it forgets with only its own spans (0.155783 vs. 0.233120) — at better plasticity (0.542204 vs. 0.517594) — while token-shuffled A-spans do not beat B’s own private spans (0.317260, against 0.375721 for no replay at all): the benefit is content, not regularization. This is replay as a continual-learning mechanism [10] with the selection made by the learner’s own surprise rather than by a curator, and with no weight-space penalty [4] in the loop. Combined with F7: a fleet of small organisms sharing an index is a *measured* design, not an aspiration — including the striking corollary that a replica which dies and rejoins through the shared store can end *better* than one that never died.

13 Method as contribution: the adversarial ledger

Every experiment in this program was preceded by a numbered, committed prediction (P1–P42 at this writing); a scoring script audits result JSONs against the register; falsified predictions remain

in the ledger with the numbers that killed them. The register is immutable in its P-numbers: a prediction is amended only before its data exists, with the reason on record, and never after.

Of the scored predictions, roughly one third were falsified or partial — including the program’s own favorite hypotheses. Four are load-bearing enough to name with their killing numbers.

- **The width law (P42, P56, §5).** “The gate law grows with width” held at two points and died at the third: 0.9729, 0.9953, 0.9777. The post-mortem then killed the post-mortem: the third point had run at a different q (a runner default), and at fixed dose the curve crosses 1.0 at $d=1024$ — while the registered “efficiency grows with width” (P56a) died cleanly at fixed dose, $0.3696 \rightarrow 0.3305$.
- **Replay cost (P39a, P39c, §11).** Registered thresholds <3.0 and <1.0 ; measured 7.085/5.603 and 2.219/2.039 on two machines, with the shared-core explanation refuted and the $2\times$ replay-cost mechanism located.
- **Relational rank (P34).** Registered: the phase lifts per-channel binding rank to $\geq 2\times$ the scalar’s. Measured on a 4-arm $\times$ 5-load $\times$ 4-seed sweep — the attention control solves every load (test recall 0.9888 to 0.9997, so the harness is sound and the task solvable), while both state arms sit at chance from $K=4$ on (chance 0.015625). The registered relational criterion had exactly one eligible cell, at $K=2$, where the ratio is 0.371 — the opposite sign of the prediction — and the cliff ratio is 1.0 against a registered ≥ 2 . The rank check itself is reported as *insufficient data* rather than as a pass or a fail: no load clears $3\times$ chance for the phase arm with the required two ignited seeds, so no cell was eligible to score. The dominating phenomenon is a different one, and it is a finding: phase ignition collapses with load (3/4 ignited seeds at $K=2$, 0/4 by $K=16$). Whatever the phase might pay at high load, training reliability dies first.
- **The one-parameter rent law (P32).** Falsified *and then explained* by a 16-cell rent map measured the same night (Figure 8).

Two method-level findings emerged from the discipline itself: **training-dynamics sensitivity to floating-point reduction order near ignition boundaries** (thread count changes *whether* a model ignites, not just speed — 6/6 seeds ignite in one regime, 0/4 in another), and the resulting rule that instruments must reproduce their reference regime. A third, from §11: an instrument must reproduce its reference *cadence* as well, or it measures overhead arithmetic rather than the property under test. We offer the ledger pattern itself as a transferable practice: a vision paper whose every claim is either a committed measurement or an explicitly open registered question.

14 The reality prediction engine (the road this opens)

The foundations compose into something none of them states alone. A system that (i) lives indefinitely on present streams at constant memory (F2, F6) — measured to seven billion tokens (§7), (ii) controls all plasticity from its own prediction error (F1) — in the spirit of temporal-difference learning’s use of prediction error as a teaching signal [9] and predictive coding’s use of prediction error as the currency of cortical inference [7], (iii) can *deposit* a prediction about its own future in a persistence mechanism built for exactly that (F3’s carrier; measured to survive silence, surgery, and weight updates) — the world-model commitment articulated for autonomous machine intelligence [5], (iv) scores the deposit when the future arrives — a loop only a *living* system can close, because a batch model never experiences its own future — and (v) exists as a portable, forkable, shardable,

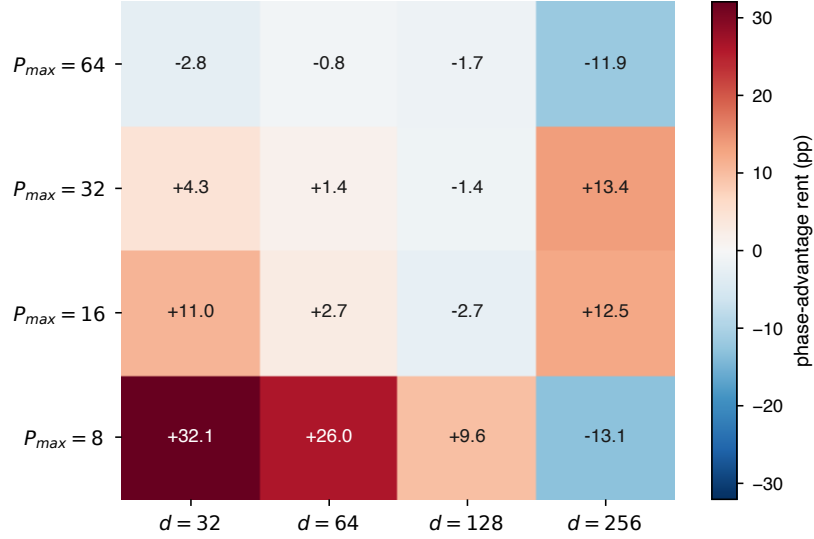


Figure 8: The 16-cell rent map: phase-advantage rent (percentage points) as a function of the recall load $P_{\max}$ and width d_{model} . The one-parameter rent law that motivated a single scalar prediction is falsified by this map’s structure; the map explains, cell by cell, where the phase advantage on real text holds and where it does not. Source: `results/holo_rent_map.json`.

collectively- remembering asset (F7, §10): this is a **reality prediction engine**. Its scaling axis is not parameters but *scope* — more streams, more horizons, more organisms sharing more index. The first horizon-ladder experiment (early-warning under distribution shift; the long-horizon head as gating teacher) is registered (P37) and running at this draft’s commit.

15 Limitations, stated plainly

Scale: $d_{\text{model}} \leq 512$, WT-2-derived vocabularies; no claim of frontier language quality is made or implied, and the width curve (§5) measures selection efficiency at a fixed token budget, not model quality. The lifetime organism’s constancy claim is about resources, not about learning: at 1.7M parameters it is at its capacity floor. Cross-ISA parity is measured at equal chunk counts only, and multi-cycle parity needs rate-matched machines (§10). Catch-up replay costs $\sim 2\times$ live streaming, which bounds continuous replication (§11). Gradient exactness requires a warmup overlap past the receptive field; with zero overlap the relative error reaches $2e-1$ (§6). The phase mechanism’s ignition is fragile (seed-, load-, and reduction-order-dependent) — a measured fact that bounds current practice and is itself a finding. The injection result is provenance-limited (stale index era). Wall-clock throughput numbers are never claimed (host contamination); all axes are token-based. The rank-1 anchor’s generating script is lost (artifact-backed, reconstruction leads documented) — reported as reproduction debt.

16 Invitation

Everything here reruns from the repository: the harnesses, the exact configurations, the raw JSONs, the ledger, and the scoring script. The organism that produced §4 can be resumed, forked, migrated, or extended by anyone — that is what F7 is for; the organism of §7 was still streaming when this draft was compiled.

Related work: predictive coding and JEPA (the vision lineage) [7, 5], TD-learning (multi-horizon error) [9], replay and weight-space penalties in continual learning [10, 4], zoology/MQAR (recall instruments) [1], Mamba/S5/LRU (the operator family) [3, 8, 6], VM live migration (the systems analogy) [2].

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